

Single-Level-Inheritance :--

Question 1 (Zepto – Grocery Delivery)

Parent Class: Order

- **Instance variables:** order_id, items, amount (initialize inside __init__)
- **Method:**
 - show_order() →
 - ✦ Create a local variable tax = 50
 - ✦ Print all order details + tax

Child Class: Delivery

- **Method:**
 - show_delivery() →
 - ✦ First, call parent method show_order()
 - ✦ Then print delivery status

Object Creation

- Create an object of Delivery with all details.
- Call show_delivery() → this will show both **order (parent) + delivery (child)**.

class order:

```
def __init__(self,order_id,items,amount):
```

```
    self.order_id=order_id
```

```
    self.items=items
```

```
    self.amount=amount
```

```
def showorder(self):
```

```
    tax=50
```

```
    print(f"order id --{self.order_id}")
```

```
    print(f"items--{self.items}")
```

```
    print(f"amount{self.amount}")
```

```
    print(f"tax--₹{self.amount+tax}")
```

```
class delivery(order):
```

```
def deliver(self):  
    super().showorder()  
    print("out of delivery")  
obj1=delivery("1","chicken biryani",400)  
obj1.deliver()  
o/p:  
order id --1  
items--chicken biryani  
amount400  
tax--₹450  
out of delivery
```

Question 2 (Amazon – E-commerce)

Parent Class: Product

- **Instance variables:** name, price, category
- **Method:**
 - show_product() →
 - ✦ Create a local variable platform = "Amazon"
 - ✦ Print product details + platform

Child Class: DiscountedProduct

- **Method:**
 - show_discount(discount_percentage) →
 - ✦ Call show_product() from parent
 - ✦ Create a local variable final_price = price - (price * discount_percentage / 100)
 - ✦ Print discount details + final price

Object Creation

- Create an object of DiscountedProduct.
- Call show_discount() → this will show **product details + discounted price**.

```

class product:

    def __init__(self,name,price,category):

        self.name=name

        self.price=price

        self.category=category

    def show_product(self):

        platform = "Amazon"

        print("----product details----")

        print(f"product name:--{self.name}")

        print(f"product price:--₹{self.price}")

        print(f"product category:--{self.category}")

        print(f"platform{platform}")

class DiscountedProduct(product):

    # def __init__(self):

    #     pass

    def show_discount(self, discount_percentage):

        self.show_product()

        final_price = self.price-(self.price * discount_percentage/100)

        print(f"discount:--{discount_percentage}%")

        print(f"final_price {discount_percentage+final_price}")

obj1=DiscountedProduct("iPhone 15",79999,"Electronics")

obj2=DiscountedProduct("Levi's Jeans",2499,"fashion")

obj3=DiscountedProduct("Philips Mixer Grinder",3499,"Home & Kitchen")

# obj1.show_product()

# obj2.show_product()

# obj3.show_product()

```

obj1.show_discount(10)

obj2.show_discount(20)

obj3.show_discount(30)

o/p:

----product details----

product name:--iPhone 15

product price:--₹79999

product category:--Electronics

platformAmazon

discount:--10%

final_price 72009.1

----product details----

product name:--Levi's Jeans

product price:--₹2499

product category:--fashion

platformAmazon

discount:--20%

final_price 2019.2

----product details----

product name:--Philips Mixer Grinder

product price:--₹3499

product category:--Home & Kitchen

platformAmazon

discount:--30%

final_price 2479.3

Question 3 (Uber – Ride Booking)

Parent Class: Ride

- **Instance variables:** ride_id, pickup, drop
- **Method:**
 - show Ride() →
 - ✦ Create a local variable distance = 12
 - ✦ Print ride details + distance

Child Class: Driver

- **Method:**
 - show_driver() →
 - ✦ Call show Ride() from parent
 - ✦ Print driver status

Object Creation

- Create an object of Driver.
- Call show_driver() → this will show **ride (parent) + driver (child)**.

class ride:

```
def __init__(self,ride_id,pickup,drop):
```

```
    self.ride_id=ride_id
```

```
    self.pickup=pickup
```

```
    self.drop=drop
```

```
def show Ride(self):
```

```
    distance=12
```

```
    print("-----ride details-----")
```

```
    print(f"ride id{self.ride_id}")
```

```
    print(f"pickup{self.pickup}")
```

```
    print(f"drop point{self.drop}")
```

```
    print(f"distance--{distance}km")
```

class driver(ride):

```
def show_driver(self):
```

```
self.show_ride()

print("driver status:available")
```

```
uber1=driver(101,"hyderabad","secunderabad")
uber2=driver(102,"delhi","pune")
uber3=driver(103,"vijayawada","narsapur")
uber1.show_driver()
uber2.show_driver()
uber3.show_driver()
```

o/p:

-----ride details-----

ride id101

pickuphyderabad

drop pointsecunderabad

distance--12km

driver status:available

-----ride details-----

ride id102

pickupdelhi

drop pointpune

distance--12km

driver status:available

-----ride details-----

ride id103

pickupvijayawada

drop pointnarsapur

distance--12km

driver status:available

In all 3:

- **Instance variables** → defined only in **parent __init__**.
- **Local variables** → exist only inside methods.
- **Parent methods** → called from child methods when needed.
- **Objects** → always created from **child class** (so it can use both parent + child methods).